BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . WEEK 19 . MODULE 16 OF 17

Module 16: Urinary System and Fluid Balance

Topics: Kidney Anatomy and Glomerular Filtration, Tubular Function and Urine Concentration, Fluid, Electrolyte, and Acid-Base Balance

TOPIC 1 OF 3 . WEEK 19 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Kidney Anatomy and Glomerular Filtration

From renal cortex to nephron to filtration barrier: how urine starts.

The Science

The kidney's gross anatomy is layered: outer cortex, inner medulla (with renal pyramids), drain to the renal pelvis to the ureter. Each kidney has about a million nephrons. Blood path: renal artery to interlobar arteries to arcuate to interlobular to afferent arteriole into the glomerulus, then efferent arteriole out, then peritubular capillaries (or vasa recta for juxtamedullary nephrons), and venous return.

The nephron is the functional unit. It has the renal corpuscle (glomerulus surrounded by Bowman's capsule) and the tubule (proximal convoluted tubule, loop of Henle, distal convoluted tubule, collecting duct). Cortical nephrons (85%) have short loops of Henle. Juxtamedullary nephrons (15%) have long loops that dip deep into the medulla and produce the concentration gradient that allows the kidney to make concentrated urine.

Filtration: the filtration barrier has three layers (fenestrated capillary endothelium, basement membrane, podocyte slit diaphragms). Cells and large proteins are excluded; water, ions, glucose, amino acids, and small molecules pass freely. GFR (glomerular filtration rate) is about 125 mL/min, or 180 L per day. Of that, less than 1% becomes urine; the rest is reabsorbed. GFR is autoregulated by myogenic response (afferent arteriole constricts in response to stretch) and tubuloglomerular feedback (macula densa senses tubular flow).

Teaching This Topic

Before the video (drawing prompt)

Have students draw a nephron with all its tubular segments and the vascular supply, then label the filtration barrier.

Common misconceptions to address

- Students think the kidney just makes urine. It also regulates BP (RAAS), acid-base balance, calcium, EPO production, vitamin D activation.
- Glomerular filtrate is sometimes confused with urine. Filtrate is roughly the composition of plasma minus proteins; urine is what is left after reabsorption.
- Tubular handling is thought of as passive. Most is active and tightly regulated.

Order of operations (what to teach first)

Gross anatomy, then the nephron structure, then the filtration barrier and GFR, then autoregulation.

Self-test prompts

- What is the typical GFR?
- Name the three layers of the glomerular filtration barrier.
- How does the kidney maintain a stable GFR despite changes in BP?

Why This Matters to Your Patients

Proteinuria suggests damage to the filtration barrier (often at the podocyte slit diaphragm in nephrotic syndrome). Hematuria can mean glomerular (RBC casts on urinalysis) or post-glomerular (stones, infection) bleeding. Acute kidney injury (AKI) can be prerenal (poor perfusion), intrinsic (the kidney tissue is damaged), or postrenal (obstruction). Chronic kidney disease (CKD) stages are based on GFR; treatment escalates from medical management to dialysis to transplant. Every patient with deranged labs (creatinine, BUN, electrolytes, acid-base) deserves a kidney-centric reading.

TOPIC 2 OF 3 . WEEK 19 . 12 CARDS (5 DOK 1 / 3 DOK 2 / 4 DOK 3)

Tubular Function and Urine Concentration

How the tubules reshape the filtrate into a final urine, and how ADH and aldosterone fine-tune it.

The Science

The tubule reshapes the filtrate. Each segment has characteristic transport biology, and each is a drug target.

Proximal convoluted tubule (PCT): bulk reabsorption. About 65% of filtered Na, water, glucose, amino acids, and bicarbonate are reabsorbed here. Glucose has a tubular maximum (Tm); above blood glucose of about 180 mg/dL the transporters saturate and glucose spills into urine (glucosuria of diabetes). Loop of Henle: descending limb is permeable to water but not ions (water leaves, filtrate concentrates); thick ascending limb is impermeable to water but actively reabsorbs Na, K, 2Cl via the NKCC2 cotransporter (target of loop diuretics like furosemide). Net effect: a hypertonic medullary interstitium (the countercurrent multiplier).

Distal convoluted tubule (DCT): reabsorbs Na and Cl via the NCC cotransporter (target of thiazide diuretics). Late DCT and collecting duct: principal cells under aldosterone control reabsorb Na and secrete K (target of potassium-sparing diuretics like spironolactone). Intercalated cells handle acid-base by secreting H+ or bicarbonate.

Hormonal control: ADH (from posterior pituitary, rises when plasma osmolarity is high or volume is low) inserts aquaporin-2 channels in the collecting duct apical membrane; water flows out of the duct into the hypertonic medulla, producing concentrated urine. Aldosterone (from adrenal cortex) drives Na reabsorption (water follows) and K secretion in the collecting duct. ANP (from atria, in volume overload) opposes RAAS and promotes Na and water excretion.

Teaching This Topic

Before the video (drawing prompt)

Have students label a nephron with each segment's job: 'PCT does X, loop does Y, DCT does Z, collecting duct does W.' They have to commit before video.

Common misconceptions to address

- Students think the collecting duct is what concentrates the urine. The collecting duct uses the gradient that the loop of Henle established; both are required.
- Aldosterone is sometimes confused with ADH. Cue: aldosterone moves SALT (and water follows); ADH moves WATER (alone).
- Diuretics are not all the same; loop, thiazide, and K-sparing have different sites and different electrolyte profiles.

Order of operations (what to teach first)

Each tubular segment in order with its main job, then the countercurrent multiplier, then ADH and aldosterone as the fine-tuning controls.

Self-test prompts

- Which segment reabsorbs the most filtered Na?
- What does ADH do mechanistically in the collecting duct?
- How does the countercurrent multiplier set up urine concentration?

Why This Matters to Your Patients

Diabetes insipidus: ADH absent (central) or ineffective (nephrogenic). Result: large volumes of dilute urine, thirst, risk of hypernatremia. Treatment: desmopressin (synthetic ADH) or address the cause. SIADH: too much ADH. Result: small volumes of concentrated urine, water retention, dilutional hyponatremia. Furosemide blocks NKCC2 in the loop; potent diuresis with hypokalemia, hypocalcemia risk. Thiazides block NCC in the DCT; milder diuresis. Spironolactone blocks aldosterone receptor; K-sparing, used in heart failure and primary hyperaldosteronism. Nurses titrating fluids and electrolytes in CHF, cirrhosis, and renal failure are working this physiology constantly.

TOPIC 3 OF 3 . WEEK 19 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Fluid, Electrolyte, and Acid-Base Balance

Body fluid compartments, the main electrolytes, and how the body keeps pH near 7.4.

The Science

Body fluid compartments: total body water is about 60% of body weight. Intracellular fluid (ICF) is about two-thirds; extracellular fluid (ECF) is about one-third. ECF subdivides into interstitial (75%) and plasma (25%). The compartments have very different electrolyte profiles: ICF is high K and low Na, while ECF is the reverse. These differences are maintained by the Na/K ATPase and matter because shifts can be catastrophic (hyperkalemia causes lethal arrhythmias because it disrupts the membrane potential).

Electrolyte priorities for the boards and the bedside: Na sets ECF osmolarity and volume; hyponatremia and hypernatremia produce neurologic symptoms. K sets membrane excitability; hyperkalemia and hypokalemia produce cardiac and muscle symptoms. Calcium has neuromuscular effects and is regulated by PTH, calcitonin, and vitamin D. Magnesium and phosphate are quieter but important cofactors and bone components.

Acid-base: normal arterial pH is 7.35-7.45. Three lines of defense. Buffers (bicarbonate, phosphate, protein) blunt acid loads on the second-to-second scale. Respiratory compensation adjusts CO2 within minutes. Renal compensation adjusts HCO3- and H+ excretion over hours to days. Four classic disturbances: respiratory acidosis (high CO2), respiratory alkalosis (low CO2), metabolic acidosis (low HCO3- or high non-CO2 acid), metabolic alkalosis (high HCO3- or loss of acid). Pattern recognition is the entire game: pH, pCO2,

HCO_3^- read together tell the story.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the body fluid compartments as nested boxes with percentages, then label which compartment is high in K versus high in Na.

Common misconceptions to address

- Students think 'edema' is too much total water. Often it is redistribution between compartments (low albumin lets fluid leak from plasma to interstitium).
- Hyponatremia is sometimes treated as low total body Na. Often it is too much water, not too little salt.
- Acid-base is reduced to memorizing equations. The story is which system is broken and which is compensating.

Order of operations (what to teach first)

Compartments and their composition, then key electrolytes and their disturbances, then acid-base with the four-disturbance pattern.

Self-test prompts

- Which compartment holds the most water?
- What does the pattern pH 7.20, pCO_2 25, HCO_3^- 12 represent?
- Why does severe vomiting cause metabolic alkalosis?

Why This Matters to Your Patients

Diabetic ketoacidosis: ketoacids accumulate, HCO_3^- is consumed buffering them, pH drops, the patient hyperventilates (Kussmaul) to drop CO_2 in compensation. Vomiting: loss of gastric H^+ causes metabolic alkalosis; volume loss adds aldosterone-driven K loss. COPD: chronic CO_2 retention with renal compensation (high HCO_3^-) and chronic compensated respiratory acidosis. Hyperkalemia (renal failure, K-sparing diuretics, Addison) is a true emergency because it causes peaked T waves and ventricular arrhythmias. Hyponatremia in elderly patients on thiazides is common and can cause seizures. Every set of labs has a fluid-and-electrolyte story; nurses who can read it catch problems early.